# Project Design Phase-II
# Technology Stack (Architecture & Stack)

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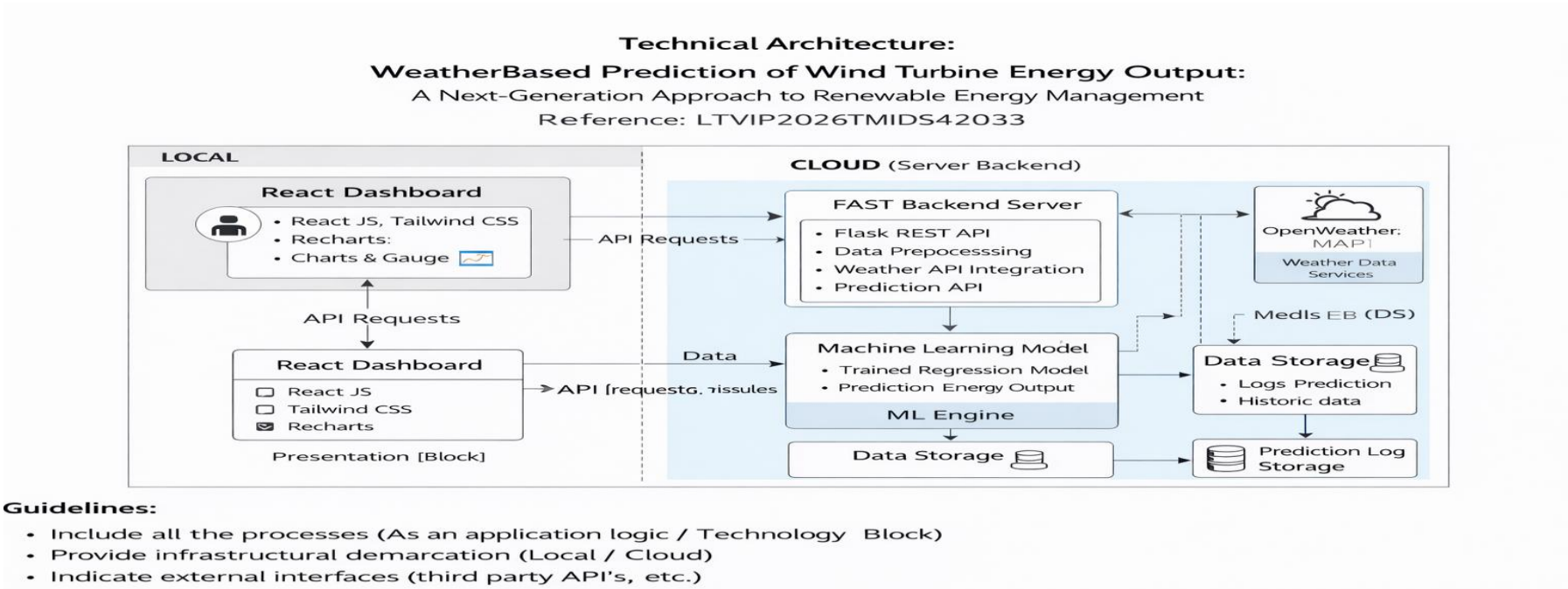
| Date | 22 February 2026 |
| Team ID | LTVIP2026TMIDS42033 |
| Project Name | **Weather-Based Prediction of Wind Turbine Energy Output: A Next-Generation Approach to Renewable Energy Management** |
| Maximum Marks | 4 Marks |

**Technical Architecture:**

The Deliverable shall include the architectural diagram as below and the information as per the table1 & table 2

**Example:**




**Guidelines:**

- Include all the processes (As an application logic / Technology Block)
- Provide infrastructural demarcation (Local / Cloud)
- Indicate external interfaces (third party API's, etc.)

**Table-1 : Components & Technologies:**

| S.No | Component | Description | Technology |
|---|---|---|---|
| 1. | User Interface | Web-based dashboard | React, TypeScript |
| 2. | Application Logic | API & ML integration | Python Flask |
| 3. | Weather API | Real-time weather data | OpenWeatherMap API |
| 4. | ML Model | Predict energy output | Scikit-learn / XGBoost |
| 5. | Database | Dataset storage | CSV / Local Storage |
| 6. | Visualization | Charts & analytics | Recharts |
| 7. | Infrastructure | Local Deployment | Windows Localhost |

**Table-2: Application Characteristics:**

| S.No | Characteristics | Description | Technology |
|---|---|---|---|
| 1. | Open-Source Frameworks | Frontend & ML libraries | React, Flask, Scikit-learn |
| 2. | Security Implementations | API Key protection | .env configuration |
| 3. | Scalable Architecture | 3-tier design | REST-based structure |
| 4. | Availability | Runs locally & cloud deployable | Docker (Future) |
| 5. | Performance | Fast API calls & prediction | Optimized ML model |

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